

HR Analytics — Employee Attrition Prediction

Final Project Report

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1. Introduction

Employee attrition is one of the most critical challenges facing modern organizations. When employees resign, companies incur significant costs related to recruitment, onboarding, and lost productivity. This project was undertaken to analyze employee attrition patterns using data science techniques and provide actionable insights to help HR teams reduce voluntary turnover. The IBM HR Analytics Employee Attrition dataset, containing 1,470 employee records across 35 attributes, was used as the foundation for this end-to-end HR analytics solution.

2. Abstract

This project applies Exploratory Data Analysis (EDA) and Machine Learning to understand and predict employee attrition. Key findings reveal that overtime work, low monthly income, young employee age (28-35), low job satisfaction, and lack of promotions are the strongest drivers of attrition. A Random Forest Classifier was built achieving 88.1% accuracy, with Monthly Income, OverTime, and Age identified as the top three predictors. A 3-page interactive Power BI dashboard was created for HR managers, along with a detailed attrition prevention PDF containing six data-backed recommendations to reduce the current 16.12% attrition rate.

3. Tools Used

Tool	Purpose
Python (Pandas, NumPy)	Data loading, cleaning and manipulation
Matplotlib & Seaborn	EDA visualizations and statistical charts
Scikit-learn	Random Forest model, metrics and feature importance
Power BI Desktop	3-page interactive dashboard with KPIs and slicers
DAX	Custom measures — Attrition Count, Rate, Avg Income
ReportLab (Python)	PDF report generation
GitHub + VS Code	Version control and development environment
Kaggle	IBM HR Analytics dataset source

4. Steps Involved

Phase 1 — Dataset & Environment Setup

Downloaded IBM HR dataset (1,470 rows, 35 columns) from Kaggle. Created project folder structure, installed Python libraries, and confirmed zero missing values. Identified and marked 4 useless columns (EmployeeCount, EmployeeNumber, Over18, StandardHours) for removal.

Phase 2 — Exploratory Data Analysis

Conducted EDA using Pandas, Matplotlib, and Seaborn. Generated 10 visualizations covering attrition by department, age, overtime, income, job satisfaction, and work-life balance. Confirmed all 5 pre-defined hypotheses and saved all charts to the outputs folder.

Phase 3 — Machine Learning Model

Built a Random Forest Classifier on 80:20 train-test split after label encoding categorical columns. Achieved 88.1% accuracy. Generated confusion matrix and feature importance chart identifying Monthly Income, OverTime, and Age as top 3 attrition predictors.

Phase 4 — Power BI Dashboard

Developed a 3-page interactive dashboard (Overview, Attrition Analysis, Employee Details) with 4 KPI cards, Department and Gender slicers, 8 charts, navigation buttons, and a Key Insights card. Created 4 DAX measures for dynamic KPI calculation.

Phase 5 — Attrition Prevention PDF

Generated an 8-page professional prevention report using Python ReportLab covering Introduction, Abstract, Key Findings, 6 Recommendations, Tools Used, Steps, and Conclusion with Future Scope.

Phase 6 — Final Project Report

Compiled this 2-page final report summarizing the entire project. All deliverables pushed to GitHub repository for submission and portfolio showcase.

5. Conclusion

This HR Analytics project successfully identified the primary drivers of employee attrition and delivered a complete data science solution comprising EDA, machine learning, an interactive dashboard, and actionable prevention recommendations. The Random Forest model achieved 88.1% accuracy, proving that attrition is predictable. The six recommendations — targeting overtime, compensation, career growth, work-life balance, and recognition — provide HR teams with a clear, evidence-based roadmap to reduce the 16.12% attrition rate. This project demonstrates that data science, when applied to human resource challenges, can transform reactive HR management into a strategic, proactive function that drives organizational success and employee wellbeing.
